

45-Day Placement Preparation Course Plan

CURRICULUM & SYLLABUS ROADMAP

Program Structure Overview

Module 1: Python Core & Advanced (22 Days) **Module 2:** SQL & Relational Databases (18 Days)
Module 3: Placement Readiness & Career Strategy (5 Days)

Part 1: Python (22 Days)

Topics & Detailed Syllabus	
Day 1	Programming Fundamentals Languages, Compilation vs Interpretation, Python Architecture, Installation, IDEs.
Day 2–3	Python Basics Variables, Data Types, Operators, Type Conversion, Input/Output.
Day 4	Conditions & Loops if/elif/else, match, for, while, break, continue, pass, patterns.
Day 5–6	Strings & Lists (Core Data Structures) Indexing, Slicing, Methods, Formatting, Comprehensions.
Day 7–8	Tuples, Sets & Dictionaries Methods, Operations, Nested Collections.
Day 9–10	Functions Parameters, Arguments, Scope, Lambda, Recursion, Built-in Functions.
Day 11–12	File Handling & Exception Handling Files, Context Manager, try/except, Custom Exceptions.
Day 13–16	Object-Oriented Programming (OOPs) Classes, Objects, Inheritance, Encapsulation, Polymorphism, Abstraction.
Day 17–20	Advanced Python Modules, Packages, Virtual Environments, Memory Management, Threads, Multiprocessing, Async.
Day 21–22	Revision, Practice & Capstone Real-world Interview Questions, DSA Practice & Mini Project.

Part 2: SQL (18 Days)

Topics & Detailed Syllabus	
Day 1	Database Fundamentals DBMS, RDBMS, SQL, Database Design, SQL Execution Flow.
Day 2–3	Database Objects CREATE, ALTER, DROP, TRUNCATE, Constraints, Data Types.
Day 4–6	SELECT Queries SELECT, WHERE, ORDER BY, DISTINCT, LIMIT, Aliases, Expressions.
Day 7–8	Operators & Clauses Comparison, Logical, IN, BETWEEN, LIKE, EXISTS, ANY, ALL.
Day 9–10	SQL Functions String, Numeric, Date, Aggregate, Conditional.
Day 11–12	Grouping GROUP BY, HAVING, Aggregations.
Day 13–14	Joins & Set Operations INNER, LEFT, RIGHT, FULL, SELF Joins & UNION.
Day 15	Subqueries & Window Functions Subqueries, ROW_NUMBER(), RANK(), DENSE_RANK().
Day 16	Views, Indexes & Transactions Views, Indexes, Transactions (COMMIT, ROLLBACK, SAVEPOINT).
Day 17	PL/SQL Fundamentals Blocks, Variables, Loops, Procedures, Functions, Triggers.
Day 18	Revision & Interview Prep Revision, Real-world Queries & Complex Interview Questions.

Part 3: Placement Preparation (5 Days)

Topics & Detailed Syllabus	
Day 1	Software Engineering & Systems Software Development Lifecycle (SDLC), Processes, Job Roles, Team Structure, System Design Basics.
Day 2	Profile Building Resume Building, ATS Optimization & GitHub Portfolio Strategy.
Day 3	Outreach & Strategy LinkedIn Optimization, Job Portals & Strategic Job Search Workflows.
Day 4	Behavioral Readiness Self Introduction, Professional Communication Skills & HR Interview Preparation.
Day 5	Evaluation & Roadmap Technical Mock Interview, Structured Feedback & Personalized Career Roadmap.